CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. To recover pure sugar from sugar solution, one uses: 1

(a) crystallisation **(b)** filtration **(c)** sublimation **(d)** decantation

2. A beam of light is visible through a mixture in a dark room. The mixture is a: 1

(a) compound **(b)** true solution
(c) pure substance **(d)** colloid or suspension

3. Evaporating 50 g of water from 200 g of 10 per cent solution makes the percentage: 1

(a) 10 per cent **(b)** 20 per cent **(c)** 13.3 per cent **(d)** 5 per cent

4. The ratio of distance moved by a component to that moved by the solvent front is the: 1

(a) density **(b)** R_f value **(c)** boiling point **(d)** solubility

5. When a saturated solution at 60 degrees Celsius is cooled to 20 degrees Celsius: 1

(a) gas is released **(b)** nothing happens
(c) more solute dissolves **(d)** crystals separate

6. Adding 100 g of water to 100 g of a 20 per cent solution changes it to: 1

(a) 10 per cent **(b)** 5 per cent **(c)** 40 per cent **(d)** 20 per cent

7. A student separates a mixture of two inks. Ink A moves 4 cm and the solvent 8 cm. The R_f of A is: 1

(a) 2 **(b)** 0.5 **(c)** 4 **(d)** 0.25

8. Which separation uses the difference in densities? 1

(a) winnowing **(b)** chromatography **(c)** sublimation **(d)** distillation

9. The solubility of a salt is 40 g per 100 g water at 30 degrees Celsius. In 250 g of water, the maximum salt that dissolves is: 1

(a) 100 g **(b)** 250 g **(c)** 40 g **(d)** 10 g

10. Impurities in iron ore can be removed by: 1

(a) sublimation **(b)** distillation
(c) magnetic separation **(d)** chromatography

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How can you show that milk is a colloid and not a solution? 2

12. Why are alloys considered mixtures although they are solid solutions? 2

13. What is the R_f value of a spot that moved 3.2 cm while the solvent moved 8.0 cm? 2

14. Why does a mixture of oil and water form two layers? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how to separate a mixture of salt, iron filings and sand. 3

16. Explain how centrifugation separates blood cells from plasma. 3

17. Describe an experiment to show that the solubility of KNO3 increases with temperature. 3

OR

Describe the setup of simple distillation and name the parts. 3

18. Explain how the layers form when a mixture of oil and water is allowed to stand. 3

OR

Name the separation technique for each: 3
(a) camphor from sand
(b) acetone from water
(c) dyes in ink.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how petroleum is refined by fractional distillation, naming five fractions and their uses. 5

OR

Explain how tests on a substance such as the boiling point or melting point can show whether it is pure. 5

20. Describe how pure water can be obtained from sea water in the laboratory, including the apparatus and the tests for purity. 5

OR

Explain how various separation methods are used in a water treatment plant for a town. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND